



Report

Roof Asset Management

FOR INFORMATION ONLY

Condition Assessment Report

Prepared for

Shaw AFB

Building:

01130-HQ Numbered AF

Shaw AFB

Sumter, SC 29152

Date inspected

07/27/2022

About Mantis Innovation Group

Mantis Innovation is a technology-driven solutions firm employing proprietary software and data analytics to improve facility performance and sustainability. Mantis' service offerings includes electricity and natural gas procurement; proactive roof, pavement, mechanical HVAC, and envelope management; design and quality assurance; demand management; energy monitoring; solar / renewable implementation; and comprehensive energy efficiency solutions. Mantis is headquartered in Houston, Texas, with over 15 additional locations across the United States from Massachusetts to Washington.

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Site Information

Site Name: 01130-HQ Numbered AF

Site Tag:

Site Address: Shaw AFB
Sumter, SC 29152

Region:

Area (Sq. Ft.): 64,459

Building Roof Access: Hatch

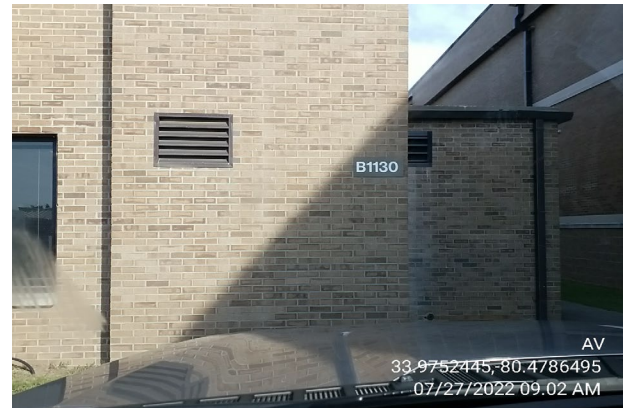
Description: Floors: Multi

Inspection Notes:

Client Contact: TBD

Phone: TBD

Email: TBD



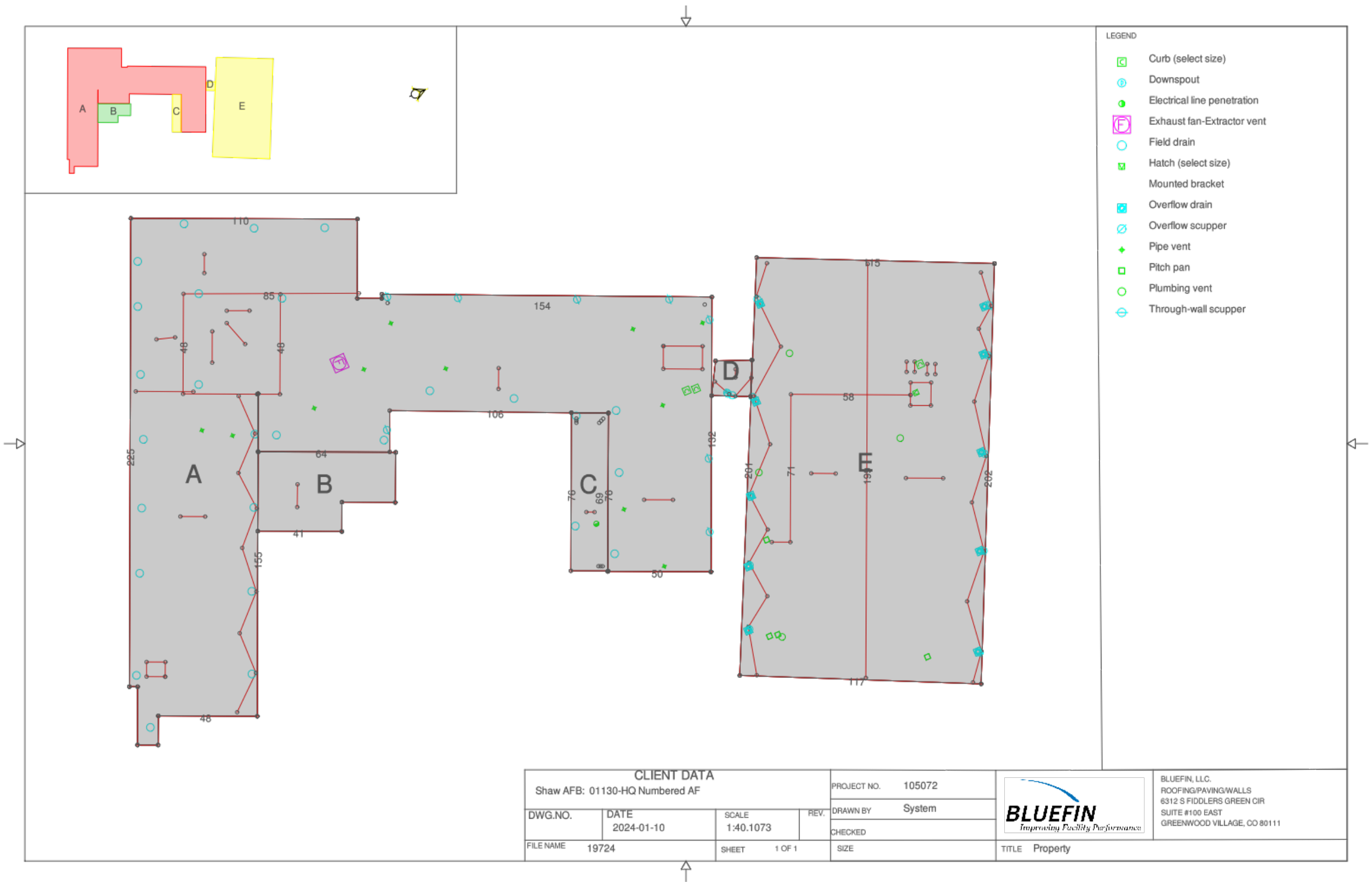
Section	Material Type	Total Area (Sq. Ft.)	Condition Score	ERSL
A	Built-up membrane	34,082	50	3
B	Metal	2,292	90	25
C	Modified bitumen	1,508	69	10
D	Modified bitumen	596	75	12
E	Modified bitumen	25,981	70	10
TOTALS (Avg.)		64,459	60	7

Recommended Repairs

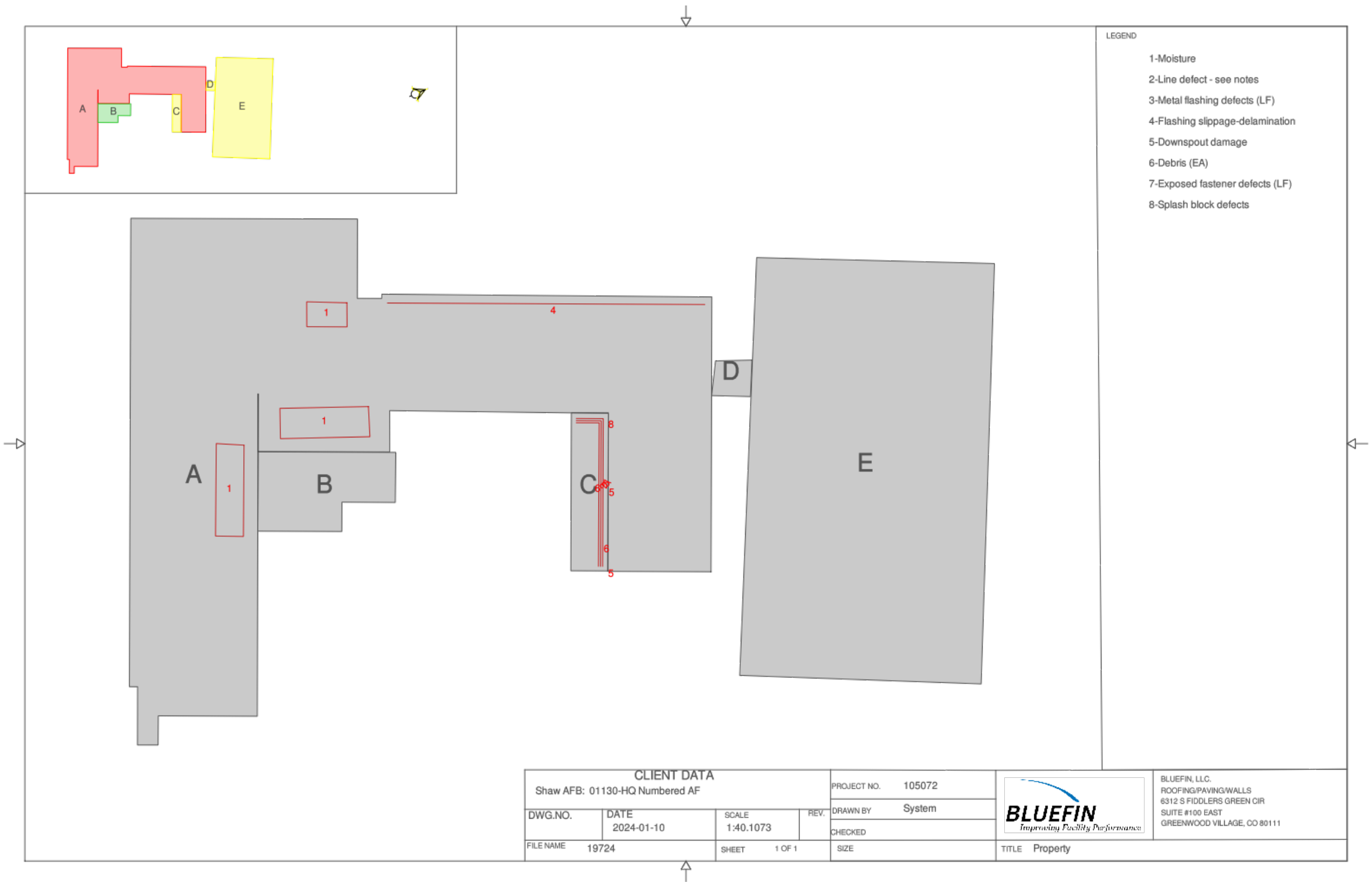
Map Link	Defect Group #	Defect	Sec	Recommendation	Defect Count (Qty U/M)	Estimate
Map	1	Moisture - Wet insulation	A	Moisture analysis required.	3 (1453 SF)	\$5,400
Map	2	Line defect - see notes - Medium severity	C	Employ a qualified Stucco / EIFS contractor to repair damaged area.	1 (80 LF)	\$2,640
Map	3	Metal flashing defects (LF) - No water penetration	C	Reposition sheet metal counter flashing to original position, secure and seal.	1 (82 LF)	\$495

<u>Map</u>	4	Flashing slippage/delamination - Delamination/slipping; no water penetration	A	Repair open seams with fabric, mastic and granules in a three- course fashion.	1 (154 LF)	\$1,650
<u>Map</u>	5	Downspout damage - Loose or missign fasteners	C	Reinstall downspout to original position and properly anchor to wall system.	2 (2 EA)	\$660
<u>Map</u>	6	Debris (EA) - Loose or light weight debris	C	Remove materials / equipment / debris and properly dispose / store offsite.	2 (2 EA)	\$330
<u>Map</u>	7	Exposed fastener defects (LF) - Loose fasteners (20-50 total)	C	Remove and replace loose / missing / corroded fasteners with new proper sized grommets fasteners.	1 (84 LF)	\$245
<u>Map</u>	8	Splash block defects - Damaged splash block	C	Supply and install splash block at downspout outlet.	1 (1 EA)	\$185
BUILDING TOTAL					12	\$11,605

Inventory Plan



Defect Plan



Section Summary - A



Section ID:	A	Type: Roof: Built-up membrane
Area:	34,082 SF	Layers:
Year Installed:	1998	● Surface:
Height:	45 FT	○ Surface: Aggregate
Sensitivity:	Low	● Membrane:
Warranty Expiration:	n/a	○ Adhesive: Asphalt
Warranty Type:	n/a	○ Adhesive: Coal tar
Condition Score:	50 (Unsatisfactory)	● Insulation:
		○ Type: Perlite cover board
		○ Thickness: See notes
		● Deck:
		○ Type: Cast in-place concrete
Notes: Prior RCS=58; prior ERSL React/Proact=6/0. Prior RCS=53; prior ERSL React/Proact=2/0. building scheduled for complete renovation. Did not remove copious debris as per Tsgt Maldonado. Much abandoned equip to remove before re-roof. Few active leaks due to concrete deck		

Section A Inventory

Inventory	Attributes	Quantity	UOM
Roof assembly: BUR surface	Surface: Aggregate	34,032	SF
Roof assembly: BUR membrane	Adhesive: Asphalt; Adhesive: Coal tar	34,032	SF
Roof assembly: BUR insulation	Type: Perlite cover board; Thickness: See notes	34,032	SF
Roof assembly: BUR deck	Type: Cast in-place concrete	34,032	SF
Perimeter: Parapet membrane flashing	Type: Composite; Attachment: Adhered; Stretch out: 12"	50	LF
Perimeter: Parapet membrane flashing	Type: Composite; Attachment: Adhered	938	LF
Perimeter: Parapet counter flashing	Type: Reglet - 1 piece	938	LF
Perimeter: Parapet counter flashing	Type: Surface mount	50	LF
Perimeter: Gravel stop		294	LF
Perimeter: Coping cap	Material: Painted steel; Width: 16"	988	LF
Perimeter: Segment	Type: Parapet	988	LF
Perimeter: Segment	Type: Edge	294	LF
Interior perimeter: Wall	Type: Block - CMU; Height: 8'	32	LF
Interior perimeter: Wall membrane flashing	Type: Roof; Attachment: Adhered	92	LF
Interior perimeter: Segment (interior)	Type: Joint	256	LF
Interior perimeter: Segment (interior)	Type: Wall	92	LF
Interior perimeter: Control joint	Width: 6"; Type: With cap; Material: Galvanized steel	228	LF
Penetrations: Pipe vent	Size: 1.50"; Material: Cast iron; Flashing: Membrane	11	EA
Penetrations: Curb (select size)	Flashing: Membrane; Base height: 18"; Length: 2'; Width: 2'; Material: Wood	2	EA
Equipment: Exhaust fan/Extractor vent	Type: Mechanized; Support: Curbed; Base height: See notes; Length: See notes; Width: See notes; Unit height: See notes	1	EA
Drainage: Field drain	Size: < 4"; Material: Aluminum	25	EA
Drainage: Through-wall scupper		6	EA
Drainage: Through-wall scupper	Type: Extruded polystyrene foam; Width: 7"; Height: See notes; Length: See notes; Material: Aluminum; Downspout: Yes	1	EA

Drainage: Through-wall scupper	Width: See notes; Height: See notes; Length: See notes; Material: See notes; Finish: See notes; Downspout: Yes	1	EA
Drainage: Cricket (measure)		165	LF
Drainage: Slope	Pitch: .25 : 12	93	LF
Assorted/Miscellaneous: Mounted bracket	Size: 4' x 4'	2	EA

Section Summary - B



Section ID:	B	Type: Roof: Metal
Area:	2,292 SF	Layers: ● Metal surface: ○ Material: Steel ○ Type: Structural panel ○ Profile: Standing seam ○ Color: Beige ○ Coating: Kynar ○ Shape: Shed
Year Installed:	2012	
Height:	25 FT	
Sensitivity:	Low	
Warranty Expiration:	n/a	
Warranty Type:	n/a	
Condition Score:	90 (Adequate)	
Notes: Prior RCS=95; prior ERSL React/Proact=28/0. Prior RCS=95; prior ERSL React/Proact=25/0.		

Section B Inventory

Inventory	Attributes	Quantity	UOM
Roof assembly: Metal roof surface	Type: Structural panel; Material: Steel; Color: Beige; Coating: Kynar; Profile: Standing seam; Shape: Shed	2,175	SF
Perimeter: Wall membrane flashing	Type: Roof; Attachment: Fastened	67	LF
Perimeter: Raised edge		38	LF
Perimeter: Raised edge	Gutter: No	38	LF
Perimeter: Drip edge	Gutter: No	67	LF
Perimeter: Segment	Type: Wall	67	LF
Perimeter: Segment	Type: Edge	143	LF
Drainage: Slope	Pitch: See notes	11	LF

Section Summary - C



Section ID:	C	Type: Roof: Modified bitumen
Area:	1,508 SF	Layers: ● Surface: ○ Color: White ○ Surface: Aluminum ● Membrane: ○ Adhesive: Asphalt ● Surface: ○ Surface: Aggregate
Year Installed:	2004	
Height:	20 FT	
Sensitivity:	Low	
Warranty Expiration:	n/a	
Warranty Type:	n/a	
Condition Score:	69 (Degraded)	
Notes: Prior RCS=80; prior ERSL React/Proact=16/0. Prior RCS=70; prior ERSL React/Proact=6/0.		

Section C Inventory

Inventory	Attributes	Quantity	UOM
Roof assembly: BUR surface	Surface: Aggregate	1,367	SF
Roof assembly: BUR membrane	Adhesive: Asphalt	1,367	SF
Roof assembly: Mod bit surface	Surface: Aluminum; Color: White	1,367	SF
Perimeter: Wall	Type: EIFS; Height: 12'; Screen: No; Wall substrate: Plywood	94	LF
Perimeter: Wall membrane flashing	Type: Roof; Attachment: Adhered; Stretch out: 18"; Wall height: 1'; Wall matl: Wood	94	LF
Perimeter: Wall counter flashing	Type: Surface mount	94	LF
Perimeter: Gravel stop	Rise: Low	94	LF
Perimeter: Segment	Type: Wall	94	LF
Perimeter: Segment	Type: Edge	94	LF
Penetrations: Electrical line penetration	Flashing: Pitch pocket; Size: 4"; Material: Galvanized steel	1	EA
Drainage: Field drain	Size: < 4"; Material: Cast iron w/ strainer	2	EA
Drainage: Slope	Pitch: .25 : 12	4	LF

Section Summary - D

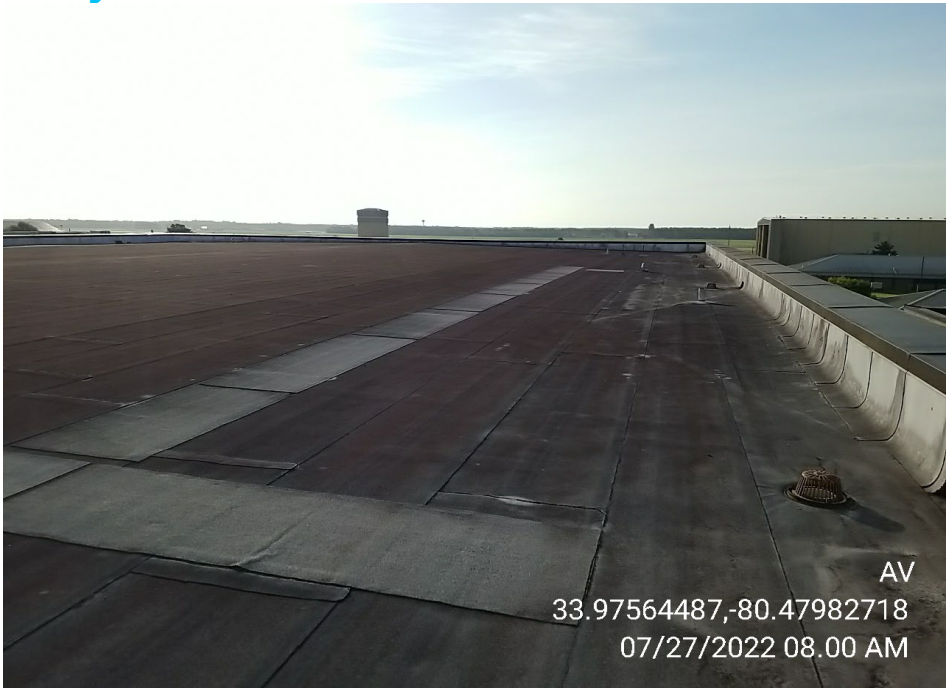


Section ID:	D	Type: Roof: Modified bitumen
Area:	596 SF	Layers: ● Surface: ○ Surface: Granular ○ Color: White ● Membrane: ○ Application: Torch ○ # of plies: 1 ● Insulation: ○ Type: Unknown
Year Installed:	2012	
Height:	50 FT	
Sensitivity:	Low	
Warranty Expiration:	n/a	
Warranty Type:	n/a	
Condition Score:	75 (Degraded)	
Notes: Prior RCS=90; prior ERS� React/Proact=17/0.		

Section D Inventory

Inventory	Attributes	Quantity	UOM
Roof assembly: Mod bit surface	Surface: Granular; Color: White	312	SF
Roof assembly: Mod bit membrane	Application: Torch; # of plies: 1	312	SF
Roof assembly: Mod bit insulation	Type: Unknown	312	SF
Perimeter: Parapet	Width: 12"; Height: 3'; Height (min): 1'	71	LF
Perimeter: Parapet membrane flashing	Type: Roof; Attachment: Adhered; Stretch out: 48"; Wall height: 3'	71	LF
Perimeter: Coping cap	Material: Painted steel; Width: 12"; Interior face: 4"; Exterior face: 5"	71	LF
Perimeter: Segment	Type: Parapet	71	LF
Drainage: Through-wall scupper	Width: 16"; Height: 8"; Length: 10"; Material: Galvanized steel; Finish: Prefinished; Downspout: Yes	1	EA
Drainage: Overflow scupper		1	EA
Drainage: Cricket (measure)		21	LF
Drainage: Downspout	Material: Steel; Type: External wall; Finish: Painted; Size: 5"; Height: 30'; Shape: Square	1	EA
Drainage: Slope	Pitch: .50 : 12	4	LF

Section Summary - E



Section ID:	E	Type: Roof: Modified bitumen
Area:	25,981 SF	Layers: ● Surface: ○ Surface: Granular ○ Color: White ● Membrane: ○ Application: Torch ○ # of plies: 1 ● Insulation: ○ Type: Unknown
Year Installed:	2012	
Height:	50 FT	
Sensitivity:	Low	
Warranty Expiration:	n/a	
Warranty Type:	n/a	
Condition Score:	70 (Degraded)	
Notes: Prior RCS=80; prior ERS� React/Proact=14/0.		

Section E Inventory

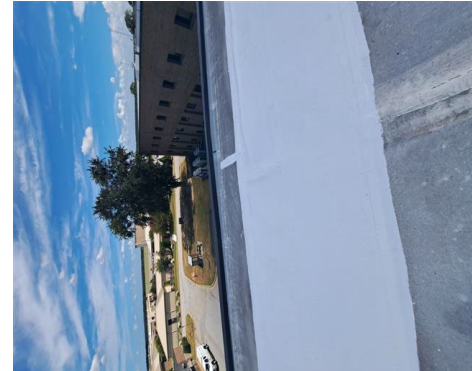
Inventory	Attributes	Quantity	UOM
Roof assembly: Mod bit surface	Surface: Granular; Color: White	23,437	SF
Roof assembly: Mod bit membrane	Application: Torch; # of plies: 1	23,437	SF
Roof assembly: Mod bit insulation	Type: Unknown	23,437	SF
Perimeter: Parapet membrane flashing	Type: Roof; Attachment: Adhered; Stretch out: 48"	636	LF
Perimeter: Coping cap	Material: Painted steel; Width: 24"	636	LF
Perimeter: Segment	Type: Parapet	636	LF
Penetrations: Plumbing vent	Size: 3"; Material: PVC; Flashing: Lead boot	2	EA
Penetrations: Plumbing vent	Size: 4"; Material: PVC; Flashing: Lead boot	2	EA
Penetrations: Hatch (select size)	Length: 3'; Width: 3'; Type: Access hatch; Base height: 8"; Material: Stainless steel; Finish: None	1	EA
Penetrations: Pitch pan	Type: Metal; Material: Stainless steel; Height: 4"; Length: 6"; Width: 6"	4	EA
Penetrations: Curb (select size)	Flashing: Membrane; Base height: 8"; Length: 1'; Width: 1'; Material: Wood	1	EA
Drainage: Overflow drain	Size: 4"	10	EA
Drainage: Through-wall scupper	Width: See notes; Height: 8"; Length: 30"; Material: Galvanized steel; Finish: Prefinished; Downspout: Yes	10	EA
Drainage: Cricket (measure)		425	LF
Drainage: Slope	Pitch: .50 : 12	30	LF
Drainage: Ridge	Ridge vent: No	199	LF
Assorted/Miscellaneous: Mod bit		180	LF
Assorted/Miscellaneous: Sleepers	Material: Wood; Size: See notes	20	LF

Defect List

Map Link	Defect Group #	Defect	Section	Recommendation	Count (Qty U/M)	Estimate Repair Cost
Map	0	Patched or repaired areas - Patch exists Status: Recorded	A	No action required at this time. Monitor and maintain.	1 (30773 SF)	\$0
Map	1	Moisture - Wet insulation Status: Proposed	A	Moisture analysis required.	3 (1453 SF)	\$5,400
Map	2	Line defect - see notes - Medium severity Status: Proposed	C	Employ a qualified Stucco / EIFS contractor to repair damaged area.	1 (80 LF)	\$2,640
Map	0	Membrane aging - Medium severity Status: Recorded	C, E, A, D	No action required at this time. Monitor and maintain.	4 (54323 SF)	\$0
Map	3	Metal flashing defects (LF) - No water penetration Status: Proposed	C	Reposition sheet metal counter flashing to original position, secure and seal.	1 (82 LF)	\$495
Map	4	Flashing slippage/delamination - Delamination/slipping; no water penetration Status: Proposed	A	Repair open seams with fabric, mastic and granules in a three-course fashion.	1 (154 LF)	\$1,650
Map	0	Flashing seam or side lap defects (EA) - Missing lap edge sealant Status: Repaired	E	BF REPAIR	1 (1 EA)	\$0
Map	0	Flashing seam or side lap defects (EA) - Open > 1/2" into seam, no water penetration Status: Repaired	D	BF REPAIR	2 (2 EA)	\$0
Map	5	Downspout damage - Loose or missign fasteners Status: Proposed	C	Reinstall downspout to original position and properly anchor to wall system.	2 (2 EA)	\$660

<u>Map</u>	0	Membrane ridges - Noticeable, felts are not exposed Status: Recorded	A	No action required at this time. Monitor and maintain.	1 (53 LF)	\$0
<u>Map</u>	0	Ponding - Ponding in membrane field Status: Recorded	A	No action required at this time. Monitor and maintain.	3 (720 SF)	\$0
<u>Map</u>	0	Metal coated surface defect - 10% of coating is deteriorated Status: Recorded	B	No action required at this time. Monitor and maintain.	1 (1876 SF)	\$0
<u>Map</u>	6	Debris (EA) - Loose or light weight debris Status: Proposed	C	Remove materials / equipment / debris and properly dispose / store offsite.	2 (2 EA)	\$330
<u>Map</u>	7	Exposed fastener defects (LF) - Loose fasteners (20-50 total) Status: Proposed	C	Remove and replace loose / missing / corroded fasteners with new proper sized grommited fasteners.	1 (84 LF)	\$245
<u>Map</u>	8	Splash block defects - Damaged splash block Status: Proposed	C	Supply and install splash block at downspout outlet.	1 (1 EA)	\$185
<u>Map</u>	0	Deteriorated sealant - Highly deteriorated Status: Repaired	D	BF REPAIR	1 (6 LF)	\$0
TOTAL					26	\$11,605

Defect Album

Defect Group #: 0**Defect:** 1 (30773 SF) Patched or repaired areas - Patch exists**Recommendation:** No action required at this time. Monitor and maintain.**Defect Group #: 1****Defect:** 3 (1453 SF) Moisture - Wet insulation**Recommendation:** Moisture analysis required.**Defect Group #: 2****Defect:** 1 (80 LF) Line defect - see notes - Medium severity**Recommendation:** Employ a qualified Stucco / EIFS contractor to repair damaged area.

Defect Group #: 0**Defect:** 4 (54323 SF) Membrane aging - Medium severity**Recommendation:** No action required at this time. Monitor and maintain.**Defect Group #: 3****Defect:** 1 (82 LF) Metal flashing defects (LF) - No water penetration**Recommendation:** Reposition sheet metal counter flashing to original position, secure and seal.**Defect Group #: 4****Defect:** 1 (154 LF) Flashing slippage/delamination - Delamination/slipping; no water penetration**Recommendation:** Repair open seams with fabric, mastic and granules in a three-course fashion.

Defect Group #: 0

Defect: 1 (1 EA) Flashing seam or side lap defects (EA) - Missing lap edge sealant

Recommendation: BF REPAIR



Defect Group #: 0

Defect: 2 (2 EA) Flashing seam or side lap defects (EA) - Open > 1/2" into seam, no water penetration

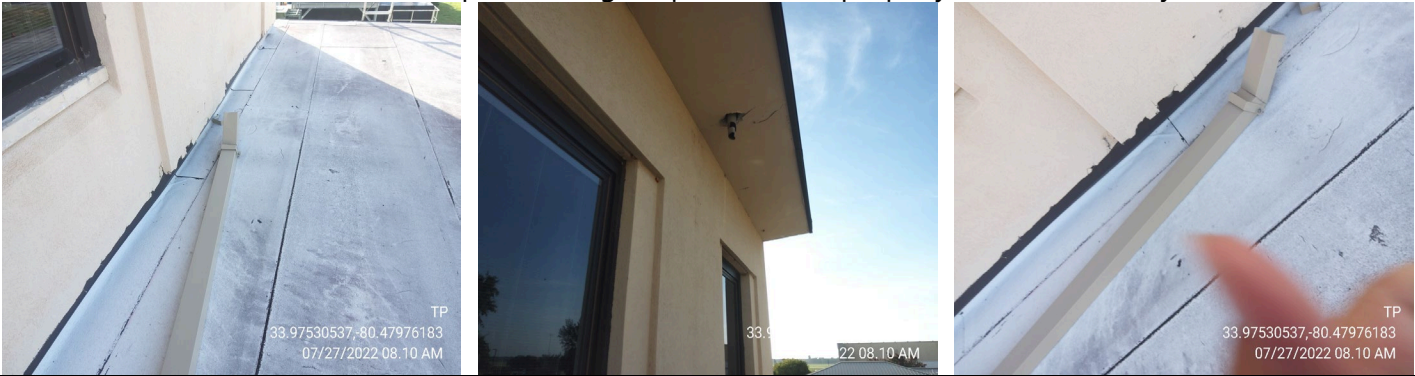
Recommendation: BF REPAIR



Defect Group #: 5

Defect: 2 (2 EA) Downspout damage - Loose or missign fasteners

Recommendation: Reinstall downspout to original position and properly anchor to wall system.



Defect Group #: 0
Defect: 1 (53 LF) Membrane ridges - Noticeable, felts are not exposed
Recommendation: No action required at this time. Monitor and maintain.



Defect Group #: 0
Defect: 3 (720 SF) Ponding - Ponding in membrane field
Recommendation: No action required at this time. Monitor and maintain.



Defect Group #: 0
Defect: 1 (1876 SF) Metal coated surface defect - 10% of coating is deteriorated
Recommendation: No action required at this time. Monitor and maintain.



Defect Group #: 6**Defect:** 2 (2 EA) Debris (EA) - Loose or light weight debris**Recommendation:** Remove materials / equipment / debris and properly dispose / store offsite.**Defect Group #: 7****Defect:** 1 (84 LF) Exposed fastener defects (LF) - Loose fasteners (20-50 total)**Recommendation:** Remove and replace loose / missing / corroded fasteners with new proper sized grommets fasteners.**Defect Group #: 8****Defect:** 1 (1 EA) Splash block defects - Damaged splash block**Recommendation:** Supply and install splash block at downspout outlet.

Defect Group #: 0
Defect: 1 (6 LF) Deteriorated sealant - Highly deteriorated
Recommendation: BF REPAIR

